

DiaTool-DPO

Enhancing Tool-Augmented LLM Dialogue
via Direct Preference Optimization

*A novel DPO-based method that models TA-LLM interactions as an MDP
with 5 dialogue states and 3 query types*

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

TA-LLMs trained only with SFT struggle with incomplete queries and out-of-scope requests

1 Missing Follow-Up

Failing to ask follow-up questions for missing information leads to incomplete tool calls.

2 Hallucinated Calls

Making hallucinated tool invocations with missing or incorrect arguments.

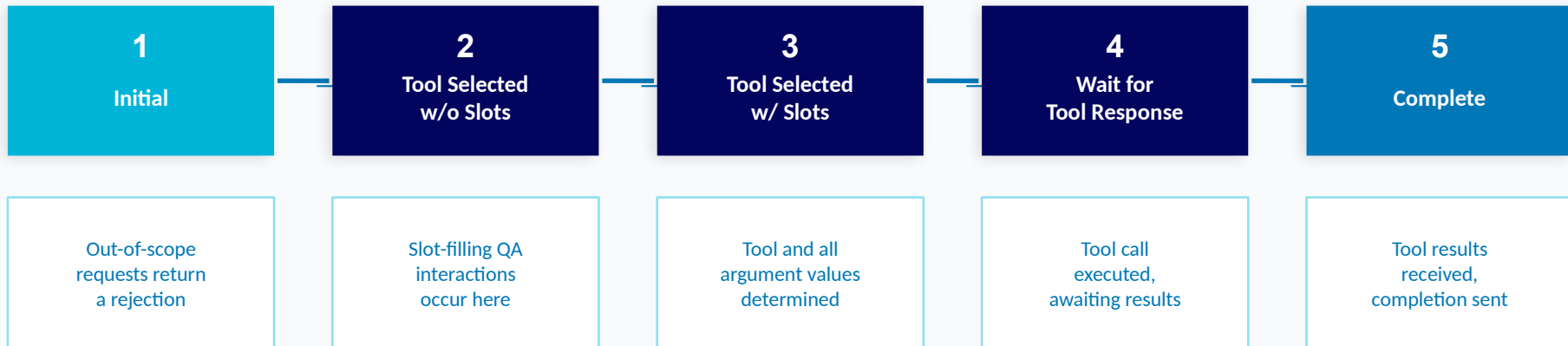
3 Failed Rejection

Failing to reject inappropriate tool calls when no suitable tools are available.

DiaTool-DPO proposes using Direct Preference Optimization to enhance dialogue capabilities by automatically constructing paired trajectory datasets of correct and incorrect dialogue flows.

Modeling TA-LLM Dialogue as a 5-State Markov Decision Process

The authors model TA-LLM interactions as an MDP with 5 internal (unobserved) states



Type 3 (Out-of-Scope): $1 \rightarrow 1$ (rejection loop)

When functionality is unavailable, the model stays in Initial state and returns rejection

Type 1 (Complete): $1 \rightarrow 3 \rightarrow 4 \rightarrow 5$ | Type 2 (Incomplete): $1 \rightarrow (2 * N) \rightarrow 3 \rightarrow 4 \rightarrow 5$

Type 2 loops through state 2 N times for slot-filling until all args are gathered

Three Query Types Define Distinct State Trajectories

Query Type	State Trajectory	Description	Example
Type 1	$1 \rightarrow 3 \rightarrow 4 \rightarrow 5$	Query contains all required arguments. No slot-filling needed.	"Find flights from NYC to L May 15"
Type 2	$1 \rightarrow (2 \times N) \rightarrow 3 \rightarrow 4 \rightarrow 5$	Query lacks arguments. Slot-filling QA repeats N times until all fields gathered.	"Book a flight" $\rightarrow$ [Which [Which date?]]...
Type 3	$1 \rightarrow 1$	Functionality unavailable. Tool call must be rejected, stays in Initial state.	"Call my bank to transfer $\rightarrow$ [Sorry, cannot help]

Key Insight

Type 1 and **Type 2** queries differ in whether slot-filling is required. **Type 3** queries require the model to recognize out-of-scope requests and appropriately reject them.

Automatic Paired Trajectory Construction Without Human Labels

Each pair teaches a specific lesson about dialogue control

Redundant Slot-Filling

Slot Hallucination

Tool Call Reject/Accept

Query	Chosen traj.	Rejected traj.	Learning	Count	Diff.
Type 1	1→3→4→5	1→2→3→4→5	Redundant slot-fill	2,089	Easy
Type 1	1→3→4→5	1→(2*N)→3→4→5	Redundant slot-fill	562	Hard
Type 1	1→3→4→5	1→(2*M)→3→4→5	Redundant slot-fill	2,530	Hard
Type 1	1→3→4→5	1→1	Tool call accept	2,090	Easy
Type 2	1→2→3→4→5	1→3→4→5	Slot hallucination	2,089	Easy
Type 2	1→(2*N)→3→4→5	1→3→4→5	Slot hallucination	562	Hard
Type 2	1→(2*N)→3→4→5	1→(2*M)→3→4→5	Slot hallucination	2,530	Hard
Type 3	Total: ~11,580 paired trajectories Easy: ~6,268 Hard: ~5,312 Automatically constructed without human annotation				
	1→1	1→3→4	Tool call reject	567	Hard

SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

Evaluation on FunctionChat-Bench: Call, Completion, Slot, Relevance metrics

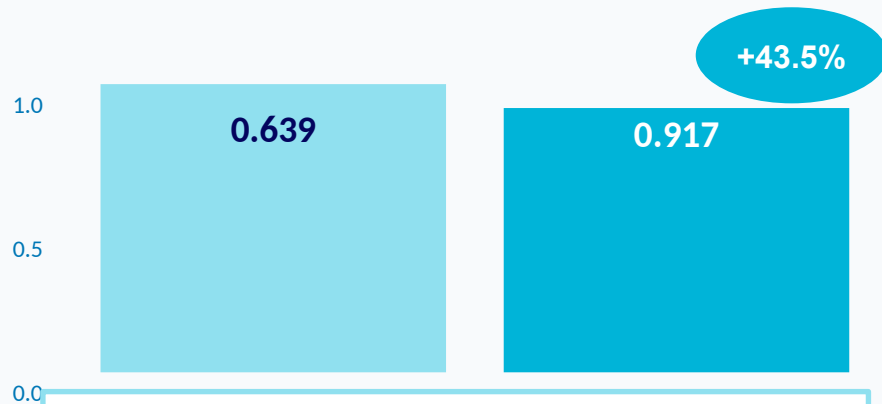
Model	Method	Call	Compl.	Slot	Rel.	Micro	Macro
Prop.-8B	DiaTool-DPO	0.314	0.700	0.833	0.609	0.575	0.614
Prop.-8B	SFT-only	0.900	0.916	0.694	0.913	0.870	0.856
Prop.-8B	SFT+DTO	0.886	0.929	0.833	0.826	0.884	0.868
Prop.-3.1B	DiaTool-DPO	0.357	0.551	0.528	0.391	0.455	0.457
Prop.-3.1B	SFT-only	0.771	0.817	0.750	0.826	0.790	0.791
Prop.-3.1B	SFT+DTO	0.743	0.871	0.833	0.826	0.765	0.818
LLaMA-3-8B	DiaTool-DPO	0.029	0.449	0.056	0.261	0.205	0.199
LLaMA-3-8B	SFT-only	0.843	0.957	0.639	0.826	0.841	0.816
							Best Macro Avg
LLaMA-3-8B	SFT+DTO	0.857	0.929	0.917	0.913	0.905	0.904

Key Finding: SFT + DiaTool-DPO consistently achieves the highest Macro Avg across all model sizes

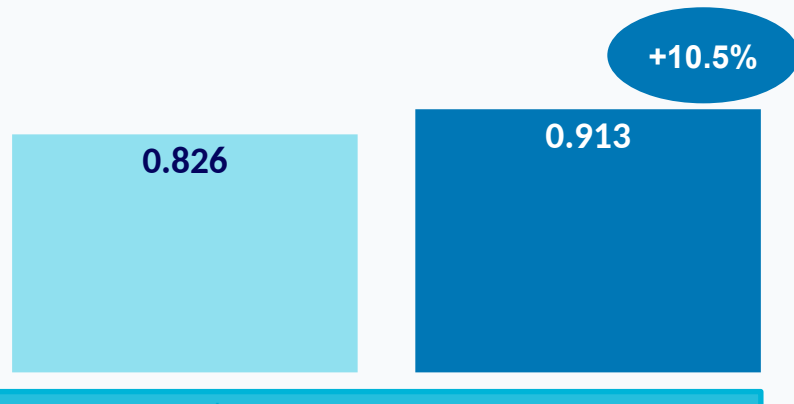
LLaMA-3-8B: Dramatic Gains with DiaTool-DPO

Slot accuracy and Relevance improvements when adding DiaTool-DPO to SFT

Slot Accuracy



Relevance



Before (SFT-only)

Slot: 0.639 | Relevance: 0.826 | Macro: 0.816

After (SFT + DiaTool-DPO)

Slot: 0.917 | Relevance: 0.913 | Macro: 0.904

Key Takeaways

RL-Based Dialogue Control for Production TA-LLMs

1

MDP Formulation Enables Systematic Trajectory Pairing

Defining 5 internal states and 3 query types allows automatic generation of chosen/rejected trajectory pairs without human annotation.

2

DiaTool-DPO Consistently Improves Macro Average

Across all three model sizes (3.1B, 8B), SFT + DiaTool-DPO achieves the best balanced performance.

3

Slot-Filling and Tool Rejection See Largest Gains

LLaMA-3-8B improves Slot accuracy by +43.5% and Relevance by +10.5% over SFT-only.

4

DPO Complements Rather Than Replaces SFT

DiaTool-DPO-only performs poorly, confirming that DPO enhances but does not substitute supervised training.

5

Approach is Language-Agnostic

Primary experiments in Korean, but method generalizes to English — opens broad applicability for production TA-LLMs.

DiaTool-DPO opens a path to GPT-4o-level conversational tool use on smaller open-source models